A low-pass filter grown from an empty embryo

Can the topology genome invent a filter that meets a mask, starting from nothing between input and output? — CLAUDE.md §4, milestone M8

kronspice simulation report · 2026-11-07 · population 300, 200 generations, seed 1

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Summary

This is the M8 milestone example: topology evolution. The embryo holds only the test fixture (after Koza et al., *Genetic Programming III*, 1999, ch. 25): a 1 V source with 1 kΩ source resistance driving node *in*, and a 1 kΩ load on node *out*. Nothing connects *in* to *out*. Evolution decides which parts exist (inductors, capacitors, resistors with E24 values), how they are wired among ground, *in*, *out* and eight internal nodes, and their values.

- **Spec met from nothing:** seed 1 grows a four-part third-order elliptic (Cauer) low-pass after 200 generations (60,000 evaluations).
- **Checked on a denser sweep:** at 200 points per decade (ten times the density evolution used) the passband stays between −6.02 and −7.12 dB (mask −5.5 to −7.5 dB) and the stopband reaches at most −46.77 dB (mask −46 dB).
- **Every seed succeeds:** seeds 1–8 all meet the mask; six of them with four parts in one of the two textbook forms of the third-order elliptic low-pass.
- **Parsimony needs time:** stopped at “spec met”, seed 1 had 15 parts at generation 50; run to 200 generations, the 0.002-per-part penalty pared it to 4.
- **Penalties, never a crash:** 15–25 % of each generation fails validation or measurement and is penalized; none panics. A run checkpointed and resumed equals the uninterrupted run.

The construction

Embryo (embryo.cir)

```
VIN vs 0 DC 0 AC 1
RS vs in 1k
RL out 0 1k
.ac dec 20 10 100k
```

With matched 1 kΩ source and load, an ideal lossless passband sits at −6.02 dB. The export sheet in *export/* shows the bare fixture: *out* at 0 V, and no automatic tearing possible because the circuit has no path to tear.

Genome (evo/encode_topology.go)

A variable-length list of parts, each with a kind, two nodes and a $\log_{10}$ value, grown into a copy of the embryo as L_g1, C_g2, …; the embryo is never modified. This is a direct netlist encoding, not Koza's developmental program. At decode, self-loops and parts with an end on an internal node nothing else touches are pruned, repeatedly: they carry no current, so every node voltage is unchanged, and without the pruning debris from mutation would turn into floating-node penalties. What remains invalid (capacitor-only islands, inductor loops) is penalized.

setting	value
terminals	in, out (plus ground)
internal nodes	8
parts	3 at start, at most 20
L	10 µH – 10 H, E24
C	100 pF – 100 µF, E24
R	1 Ω – 1 MΩ, E24
GA	population 300, 200 generations, seed 1, 2 s timeout per evaluation

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Operators. Change value (Gaussian, in decades), add, delete, rewire one end, change kind (keeping the relative position in the range), split a part in series through a fresh node, add a part in parallel; one operator per child plus a geometric number more (mean 1.43). Crossover is one-point on the part lists. New parts attach to nodes already in use plus one fresh internal node, so they mostly join the circuit.

Objectives

mask band	requirement
passband 10 Hz – 1 kHz	−7.5 dB ≤ gain ≤ −5.5 dB (matched: −6.02 dB)
stopband 3 kHz – 100 kHz	gain ≤ −46 dB (40 dB below the passband)

Fitness is the mask error (*evo.FilterMask*) plus 0.002 per part (*evo.PartCount*). The mask error is 0 when the response stays inside the mask at every sweep point and band edge; otherwise it is 0.1 + the fraction of points outside + the mean excess in dB/20, after Koza's filter scoring. The 0.1 floor means any design that leaves the mask scores worse than any design that meets it, so “fitness < 0.1” reads “spec met”, and once the mask is met evolution keeps removing parts.

Results

Seed 1: the evolved filter

```
L_g1 in  out 0.2       series inductor
C_g4 in  out 11n       across it: a notch at 1/(2π√(LC)) = 3.39 kHz
C_g2 0   in  180n      shunt capacitors at both ends
C_g3 out 0   220n
```

That is a third-order elliptic (Cauer) low-pass: a π-section with the trap across the series arm. Re-simulated from the written *best.cir* on a sweep of 200 points per decade, the passband stays between −6.02 and −7.12 dB and the stopband reaches at most −46.77 dB, inside the mask everywhere; this re-simulation is the acceptance test. *best.cir* opens in ngspice, LTspice and QSpice as it is.

Seeds 1–8 (200 generations, population 300)

seed	parts	form
1	4	series L with C across it (notch 3.4 kHz), shunt C at both ends
2	6	same idea, with two small resistors
3	4	series L with C across it, shunt C both ends
4	4	series L, shunt C, series-LC trap to ground at the input (notch 3.9 kHz)
5	6	two series L, trap across, shunt C both ends, one small resistor
6	4	series L with C across it, shunt C both ends
7	4	series L, shunt C, series-LC trap to ground
8	4	series L, shunt C, series-LC trap to ground

All eight seeds meet the mask. The four-part designs are the two textbook forms of the third-order elliptic low-pass: the trap across the series inductor, or a series-resonant trap in a shunt arm. Third order is the lowest that can meet about 1 dB ripple and 40 dB attenuation from 3× the passband edge (a Butterworth needs five reactive parts, a Chebyshev four); here it is reached from an empty fixture with standard values.

How the search behaves

- **Parsimony needs time.** With an early stop at “spec met”, seed 1 stopped at generation 50 with 15 parts; run to 200 generations, the 0.002-per-part penalty pared it to 4.
- **Penalties, never a crash.** Every decoded circuit passes *circuit.Validate* before simulation; 15–25 % of each generation fails validation or measurement and takes a penalty from the M5 classes (no-measurement < no-convergence < singular < invalid < ledger < timeout < other < panic). None panics.
- **Deterministic and resumable.** Operators draw from the per-generation PCG stream; a topology run checkpointed at generation 2 and resumed equals the uninterrupted run (*TestTopologyResume*).

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Validation

M8 criterion	status (doc/M8.md)
filter from an empty embryo meets spec	*TestFilterTopologyExample*: written best netlist re-simulated at 200 points/decade stays inside the mask (passband −6.02…−7.12 dB, stopband ≤ −46.77 dB)
every decoded circuit validated before simulation	*circuit.Validate* before any analysis
penalties, never a crash	M5 penalty classes; none panics
deterministic, resumable	*TestTopologyResume*
go test -race ./...	clean

Limits

- Koza's developmental (circuit-constructing) encoding is not implemented; the direct encoding was enough for this milestone.
- Only R, C and L are evolvable parts; transistors and sources are not.
- No multi-part local search.

Reproducing the results

```
go run ./cmd/evolve -config examples/filter_topology/config.json   # about 40 s
go test ./evo -run 'TestFilterTopologyExample|TestTopologyResume'   # the M8 acceptance tests
python examples/filter_topology/viz/make_report.py examples/filter_topology/filter_topology_report.pdf
```

Source: J. R. Koza, F. H. Bennett III, D. Andre, M. A. Keane, Genetic Programming III: Darwinian Invention and Problem Solving, Morgan Kaufmann (1999), ch. 25 — the embryo fixture, the filter mask scoring and the evolved low-pass filters this example repeats with a direct encoding.

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